

DM Module: DLA 01: Glycolysis (Required reading assignment)

OBJECTIVES for DM DLA 01: Glycolysis (Required reading assignment)

SOM. MK.I. BPM2.3. DM.2.BCHM.1084	Define glycolysis and explain its role in the generation of metabolic energy.
SOM. MK.I. BPM2.3. DM.2.BCHM.1085	List the reactions and enzymes that convert glucose into pyruvate (emphasize key enzymes and irreversible reactions)
SOM. MK.I. BPM2.3. DM.2.BCHM.1086	Describe substrate level phosphorylation and identify those reactions in glycolytic pathway
SOM. MK.I. BPM2.3. DM.2.BCHM.1087	Explain significance of lactate production in anaerobic glycolysis. Explain the further fate of lactate formed in RBC and muscle and highlight importance of the Cori cycle.
SOM. MK.I. BPM2.3. DM.2.BCHM.1088	Outline the regulation of glycolysis indicating the regulatory enzymes. Highlight the role of AMP, ATP on glycolysis.

Required Reading before DM Lecture 01:

- Lippincott's Biochemistry 9th Edition
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott's Biochemistry **Chap 8 Glycolysis** [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- **Chapter 8: Introduction to Metabolism and Glycolysis;**
 - **Sections: Glycolysis overview; Glucose transport into cells; Glycolysis reactions; Hormonal regulation; Alternate fates of pyruvate; Chapter concept map: Fig: 8.25**
- Lippincott's Biochemistry (Pentose phosphate pathway) Chapter 13; Concept map Fig 13.14; [Pentose Phosphate Pathway and Nicotinamide Adenine Dinucleotide Phosphate | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - **Sections in the Chapter:** Chapter summary: Fig 13.14